

Supply Chain Monitoring and Optimization Platform

Capstone Project — Week 5

Azure DevOps CI/CD Automation

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Submission Date: 13-05-2026	Week: Week 5

WEEK 5 – AZURE DEVOPS PIPELINE SIMULATION

Objective

In week 5 i learned the basic idea of ci/cd pipeline automation using azure devops. Since azure services were not available during implementation, the workflow was simulated in google colab using sample yaml configuration and python execution flow.

Tools Used

- Google Colab
- Python
- Yaml Pipeline Configuration

Workflow

```
Push Code
↓
Pipeline Starts
↓
Install Dependencies
↓
Run Python Script
↓
Display Output Logs
```

Sample Pipeline Yaml

```
Trigger:
- Main

Pool:
  Vmimage: 'Ubuntu-Latest'

Steps:
- Script: |
  Pip Install Pandas Numpy Pyspark
  Python Run_Pipeline.Py
  Displayname: 'Run Supply Chain Script'
```

This Yaml File Defines The Basic Pipeline Flow.

Sample Python Pipeline Script

```
print("Pipeline Started")

print("Loading Supply Chain Data")
```

```
print("Processing Data")  
  
print("Pipeline Completed Successfully")
```

This Script Simulates Execution Of A Simple Automated Pipeline.

Expected Output

- Pipeline Execution Started
- Dependencies Installed
- Python Script Executed
- Output Logs Displayed Successfully

Conclusion

This Task Helped In Understanding The Basic Workflow Of Ci/Cd Automation And How Pipelines Can Be Used To Automate Execution Of Python Scripts In Data Engineering Projects.